

**Tribhuvan University**  
**Institute of Engineering**  
**Department of Electronics and Computer Engineering**  
**Pulchowk, Nepal.**

**Title: Machine Learning and Pattern Recognition**

Course Code:

Level: M.Sc. Credit hours: 60

Lecture: 3

Program: Information and Communication

Tutorial: 1

Year:

Part:

| Examination scheme |       |                      |                                                                                       |
|--------------------|-------|----------------------|---------------------------------------------------------------------------------------|
| Internal           | Final | Examination Duration | Remarks                                                                               |
| 40                 | 60    | 3 hrs                | International assessment may include minor tests, assignments and small project works |

**Objective of the course:**

To provide a fundamental understanding of machine learning and pattern recognition and its useful application in computer vision, information security and digital imaging and printings.

**Course description:**

This course presents comprehensive introduction to several topics on pattern recognition (PR) and machine learning (ML). This course explores the application of ML in computer vision, digital imaging, biometrics and information security.

**Indicative Course contents:**

|                                                   |       |
|---------------------------------------------------|-------|
| 1 Machine Learning Basics                         | 5hrs  |
| 1.1 History of Machine Learning                   |       |
| 1.2 Definition of Learning                        |       |
| 1.2.1 Types of Natural Learning                   |       |
| 1.2.2 Need of Machine Learning                    |       |
| 1.3 Machine Learning Models                       |       |
| 2 Measures for Performance Evaluation             | 10hrs |
| 2.1 Classification accuracy                       |       |
| 2.2 Confusion matrix                              |       |
| 2.3 Misclassification costs                       |       |
| 2.4 Class probability estimation                  |       |
| 2.5 Information score                             |       |
| 2.6 Performance evaluation for two class problems |       |
| 2.6.1 Sensitivity and specificity                 |       |
| 2.6.2 ROC curve                                   |       |
| 2.6.3 Recall and precision                        |       |
| 3 Estimating Performance                          | 10hrs |

|                                                    |        |
|----------------------------------------------------|--------|
| 3.1 Reliability of quality estimations             |        |
| 3.2 Confidence interval                            |        |
| 3.3 Cross validation                               |        |
| 3.4 Data visualization                             |        |
| 4 Basics of pattern recognition                    | 5hrs   |
| 4.1 Terminology                                    |        |
| 4.2 Class                                          |        |
| 4.3 Test dataset                                   |        |
| 4.4 Training dataset                               |        |
| 4.2 Review of statistics                           |        |
| 5 Bayesian Decision Theory                         | 10hrs  |
| 5.1 Classifiers and Discriminant functions         |        |
| 5.2 Decision surfaces                              |        |
| 5.3 Normal density                                 |        |
| 5.3 Discrete features                              |        |
| 6 Dimension reduction method                       | 5hrs   |
| 6.1 Principle component analysis                   |        |
| 7 Non parametric techniques for density estimation | 5hrs   |
| 7.1 K-nearest neighbor                             |        |
| 7.2 Linear discriminant classifier- perceptron     |        |
| 8 Unsupervised learning and clustering             | 10 hrs |
| 8.1 Criterion function for clustering              |        |
| 8.2 Algorithms for clustering                      |        |
| 8.2.1 K-means                                      |        |
| 8.2.2 Hierarchical and other methods               |        |
| 8.3 Cluster validation                             |        |

### References:

1. R.O.Duda, P.E.Hart and D.G.Stork, Pattern Classification, John Wiley, 2001
2. S.Theodoridis and K.Koutroumbas, Pattern Recognition, Academic Press, 2009
3. C.M.Bishop, Pattern Recognition and Machine Learning, Springer, 2006